

TITLE:

Comprehensive Security Monitoring and Threat Detection using Wazuh

Acknowledgement

This work is the result of my own effort, dedication, and learning. I am thankful for the resources, knowledge, and opportunities that enabled me to explore, create, and bring my ideas to life. This project has been a valuable journey of self-discovery and skill enhancement, and I am proud to present it as a reflection of my hard work and passion.

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1. Introduction

In today's digital world, cybersecurity plays a crucial role in protecting systems and data from malicious activities. This project focuses on implementing a comprehensive security monitoring system using Wazuh.

Wazuh is an open-source security platform that provides threat detection, integrity monitoring, and incident response capabilities. This project demonstrates how Wazuh can be used to monitor systems, detect attacks, and analyze security events in real time.

2. Objectives

- To implement file integrity monitoring
- To detect network intrusions using Suricata
- To identify system vulnerabilities
- To detect execution of malicious commands
- To prevent SSH brute force attacks
- To analyze malicious files using VirusTotal

3. System Overview

The system consists of:

- Wazuh Manager (central server)
- Wazuh Agents (installed on monitored systems)
- Suricata IDS for network monitoring

All logs and events are collected and analyzed centrally, providing a unified security monitoring platform.

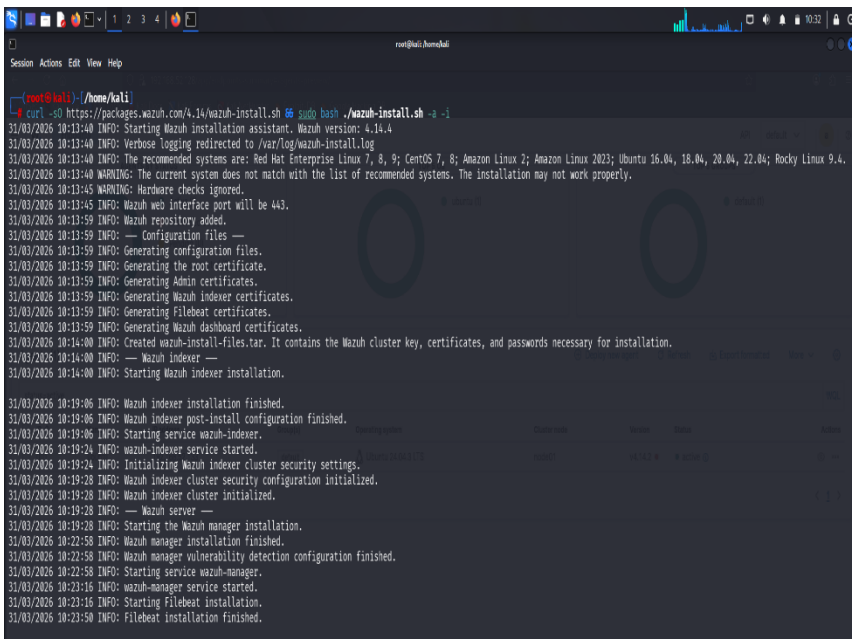
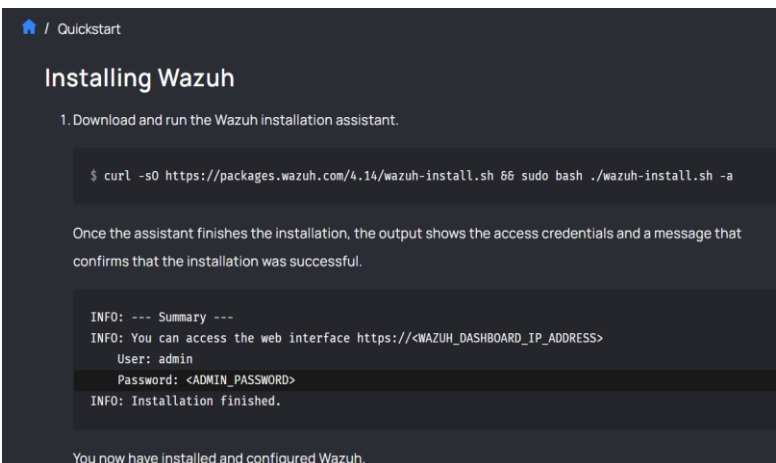
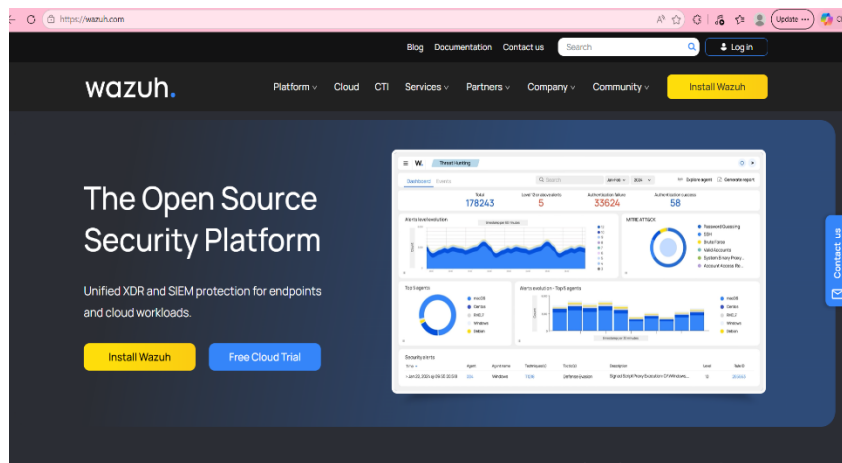
4. Methodology

The project is implemented in the following steps:

1. Install and configure Wazuh
2. Deploy agents on target systems
3. File Integrity Monitoring (FIM)
4. Integrate Suricata IDS for network monitoring
5. Detecting vulnerabilities
6. Detecting Execution of Malicious Commands
7. Detecting and Blocking SSH brute force attack
8. Detecting malicious files using Virustotal

Implementation

1. Install and configure Wazuh



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2. Deploy agents on target systems

Deploy new agent

Select the package to download and install on your system:

LINUX

☐ RPM amd64 ☐ RPM aarch64 ☒ DEB amd64 ☐ DEB aarch64

WINDOWS

☐ MSI 32/64 bits

macOS

☐ Intel ☐ Apple silicon

For additional systems and architectures, please check our [documentation](#).

Server address:

This is the address the agent uses to communicate with the server. Enter an IP address or a fully qualified domain name (FQDN).

Assign a server address

192.168.52.128

☒ Remember server address

Optional settings:

Select one or more existing groups:

Default

4 Run the following commands to download and install the agent:

```
wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.14.2-1_amd64.deb && sudo WAZUH_MANAGER='192.168.52.128' WAZUH_AGENT_NAME='ubuntu' dpkg -i ./wazuh-agent_4.14.2-1_amd64.deb
```

Requirements

- You will need administrator privileges to perform this installation.
- Shell Bash is required.

Keep in mind you need to run this command in a Shell Bash terminal.

5 Start the agent:

```
sudo systemctl daemon-reload
sudo systemctl enable wazuh-agent
sudo systemctl start wazuh-agent
```

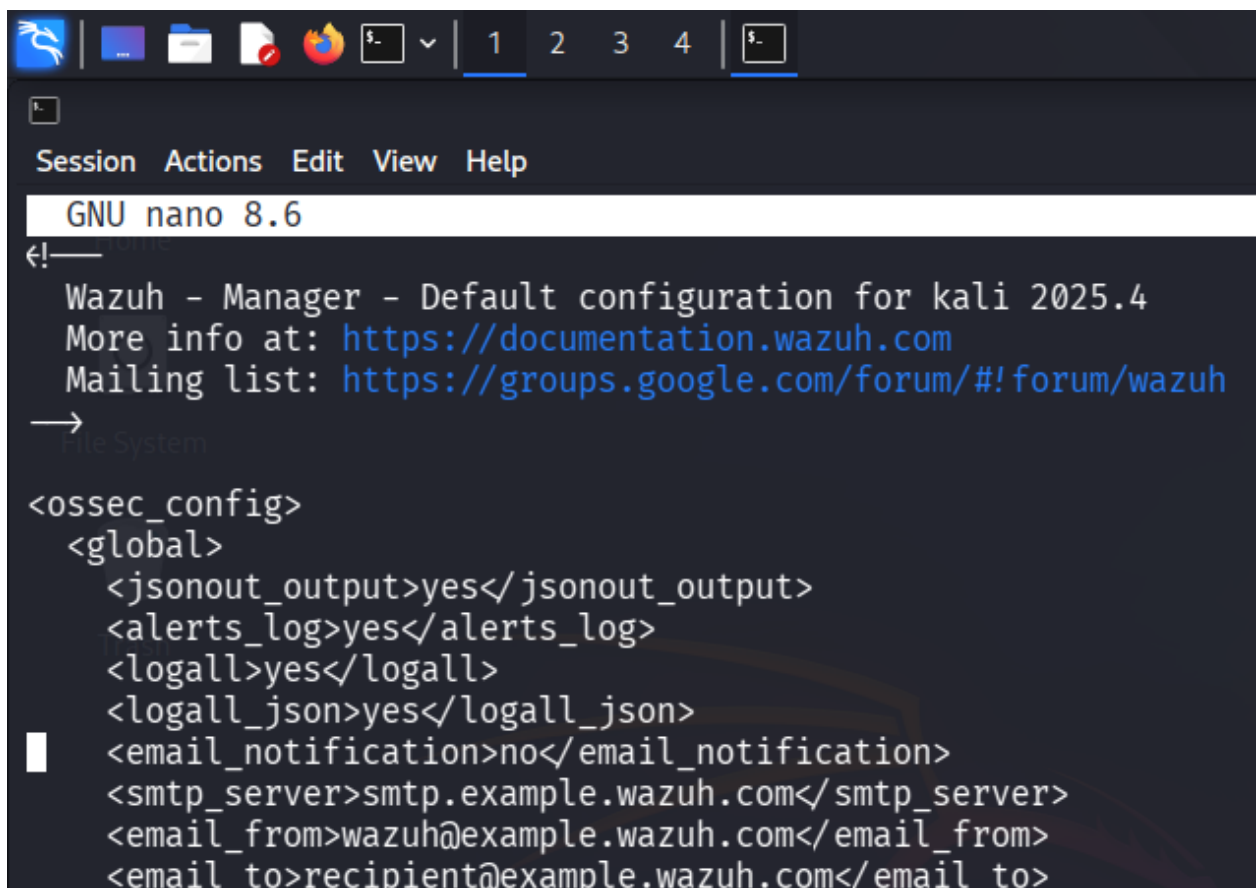
```
owais@owais-Virtual-Platform:~$ sudo su
[sudo] password for owais:
root@owais-Virtual-Platform:~# wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.14.2-1_amd64.deb && sudo WAZUH_MANAGER='192.168.52.128' WAZUH_AGENT_NAME='ubuntu' dpkg -i ./wazuh-agent_4.14.2-1_amd64.deb
2026-04-02 10:18:19 - https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.14.2-1_amd64.deb
Resolving packages.wazuh.com (packages.wazuh.com)... 99.86.39.5, 99.86.38.81, 99.86.39.23, ...
Connecting to packages.wazuh.com (packages.wazuh.com)[99.86.39.9]:443... connected.
HTTP request sent, awaiting response... 200 OK
length: 13121764 (13M) [application/vnd.debian.binary.package]
Saving to: 'wazuh-agent_4.14.2-1_amd64.deb.1'
wazuh-agent_4.14.2-1_amd64.deb.1 100%[=====] 12.51M 3.46MB/s in 3.6s
2026-04-02 10:18:23 (3.46 MB/s) - 'wazuh-agent_4.14.2-1_amd64.deb.1' saved [13121764/13121764]
(Reading database ... 193065 files and directories currently installed.)
Preparing to unpack .../wazuh-agent_4.14.2-1_amd64.deb ...
Unpacking wazuh-agent (4.14.2-1) over (4.14.2-1) ...
Setting up wazuh-agent (4.14.2-1) ...
root@owais-Virtual-Platform:~#
```

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3. File Integrity Monitoring (FIM)

- First, we've to configure the rules for file monitoring in the wazuh server (where wazuh manager is installed).
- On the server run the following command in the terminal:
"nano /var/ossec/etc/ossec.conf"
- By default, "logall" tag is off and we've to write "yes" to turn it on as shown
- Turn on "alerts_log" and "logall_json" tag as well.



The screenshot shows a terminal window with the nano text editor open. The editor is displaying the configuration file /var/ossec/etc/ossec.conf. The configuration is for Wazuh Manager, version 2025.4. The current section is <ossec_config>, and the sub-section is <global>. The following tags are being configured: <jsonout_output>yes</jsonout_output>, <alerts_log>yes</alerts_log>, <logall>yes</logall>, <logall_json>yes</logall_json>, <email_notification>no</email_notification>, <smtp_server>smtp.example.wazuh.com</smtp_server>, <email_from>wazuh@example.wazuh.com</email_from>, and <email_to>recipient@example.wazuh.com</email_to>. The cursor is positioned at the end of the <logall_json>yes</logall_json> line.

```
GNU nano 8.6
Wazuh - Manager - Default configuration for kali 2025.4
More info at: https://documentation.wazuh.com
Mailing list: https://groups.google.com/forum/#!forum/wazuh
--
File System
<ossec_config>
  <global>
    <jsonout_output>yes</jsonout_output>
    <alerts_log>yes</alerts_log>
    <logall>yes</logall>
    <logall_json>yes</logall_json>
    <email_notification>no</email_notification>
    <smtp_server>smtp.example.wazuh.com</smtp_server>
    <email_from>wazuh@example.wazuh.com</email_from>
    <email_to>recipient@example.wazuh.com</email_to>
```

- Now go to the wazuh agent system and navigate to same file as in wazuh server by running the same command "nano /var/ossec/etc/ossec.conf" in the terminal.
- By scrolling down in the file, the section <!-- File integrity monitoring --> needs to be modified.

- Add the following attribute to the section `<directories check_all="yes" report_changes="yes" realtime="yes">/root</directories>` as shown below:

```
<!-- File integrity monitoring -->
<syscheck>
  <disabled>no</disabled>

  <!-- Frequency that syscheck is executed default every 12 hours -->
  <frequency>43200</frequency>

  <scan_on_start>yes</scan_on_start>

  <!-- Directories to check (perform all possible verifications) -->
  <directories>/etc,/usr/bin,/usr/sbin</directories>
  <directories>/bin,/sbin,/boot</directories>
  <directories check_all="yes" report_changes="yes" realtime="yes">/root</directories>
```

- Save the file.
- Restart the wazuh agent (systemctl restart wazuh-agent)
- For demo let's make a file named "sample_file.txt" by entering the command "touch sample_file.txt" in the agent system.
- This file should be logged in the wazuh manager as shown:

The screenshot displays the Wazuh web interface. The top navigation bar includes 'Dashboard', 'Inventory', and 'Events'. The 'Events' tab is active, showing a search bar and filters for 'manager.name: kali', 'rule.groups: syscheck', and 'agent.id: 001'. A bar chart shows a single event at 03:00. Below the chart, a table lists the event details:

timestamp	agent.name	syscheck.path
Apr 4, 2026 @ 03:28:26.004	ubuntu	/root/sample_file.txt

The 'Document Details' panel on the right shows the JSON representation of the event:

```
{
  "_index": "wazuh-alerts-4.x-2026.04.04",
  "_type": "log",
  "agent.id": "001",
  "agent.ip": "192.168.52.133",
  "agent.name": "ubuntu",
  "decoder.name": "syscheck_new_entry",
  "full_log": "File '/root/sample_file.txt' added\nMode: realtime",
  "id": "1775267768.69197",
  "location": "syscheck",
  "manager.name": "kali",
  "rule.description": "File added to the system.",
  "rule.firedtimes": "1",
  "rule.gdpr": "II.5.1.f",
  "rule.gpg13": "4.11",
  "rule.groups": "ossec, syscheck, syscheck_entry_added, syscheck_file",
  "rule.hipaa": "164.312.c.1, 164.312.c.2",
  "rule.id": "554"
}
```

- As we can see the log is generated.
- It also shown the time, date, index, agent.id, agent.ip, full log, which rule was triggered etc.
- This completes our File integrity monitoring (FIM).

4. Integrate Suricata IDS for network monitoring

→ Installing Suricata IDS on Ubuntu agent

- Install suricata on the Ubuntu endpoint by running following commands:
“sudo add-apt-repository ppa:oisf/suricata-stable
sudo apt-get update
sudo apt-get install suricata -y”
- Navigate to “/etc/suricata” and make a directory named “rules” by using command “mkdir rules”.
- Now install the rules in this directory by following command:
“cd /tmp/ && curl -LO https://rules.emergingthreats.net/open/suricata-6.0.8/emerging.rules.tar.gz
sudo tar -xvzf emerging.rules.tar.gz && sudo mv rules/*.rules
/etc/suricata/rules/
sudo chmod 640 /etc/suricata/rules/*.rules”
- Now let’s customise the setting for suricata
 1. We need to modify file /etc/suricata/suricata.yaml and set the following variables:
“HOME_NET: "<UBUNTU_IP>"
EXTERNAL_NET: "any"

default-rule-path: /etc/suricata/rules
rule-files:
- "*.rules"

Global stats configuration
stats:
enabled: Yes

Linux high speed capture support
af-packet:
- interface: eth0”
- For the modification navigate to file by the command:
“nano /etc/suricata/suricata.yaml”
- Inside the file press ctrl+w and search “af-packet”
- Change the interface name as your interface name by running the command “ifconfig” in the terminal.
- Now search for “global stats configuration” and it should be enabled.
- Now go to “HOME_NET” and put your agents ip-address.

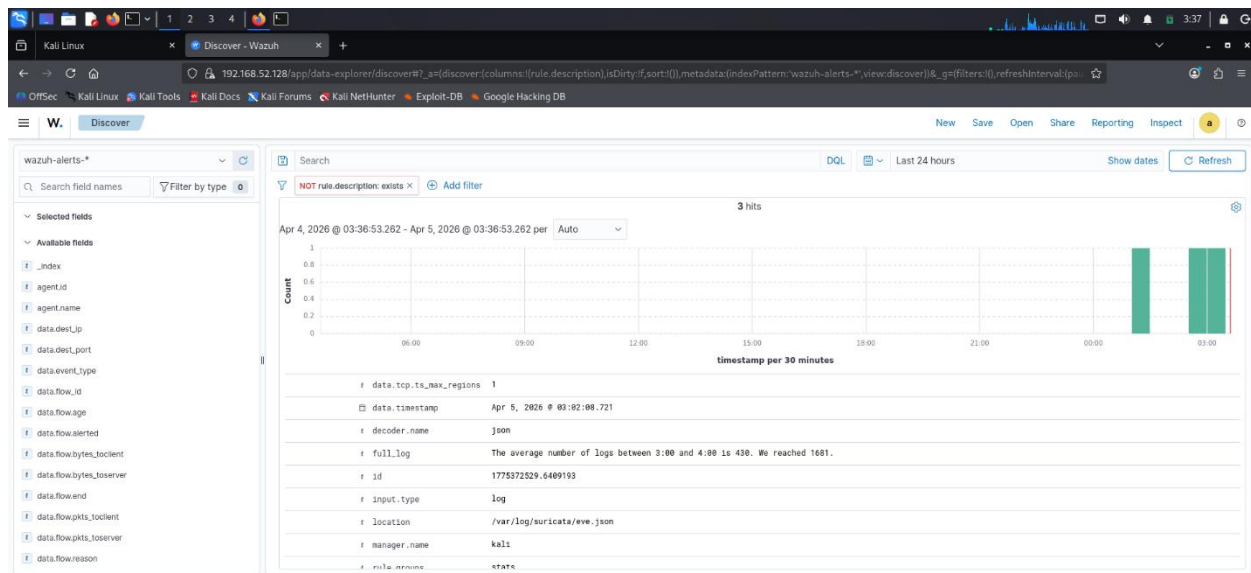
- Comment EXTERNAL_NET: “!HOME_NET” and uncomment EXTERNAL_NET: “any” as shown:

```
vars:
  # more specific is better for alert accuracy and performance
  address-groups:
    HOME_NET: "[192.168.52.133]"
    #HOME_NET: "[192.168.0.0/16]"
    #HOME_NET: "[10.0.0.0/8]"
    #HOME_NET: "[172.16.0.0/12]"
    #HOME_NET: "any"

    #EXTERNAL_NET: "!$HOME_NET"
    EXTERNAL_NET: "any"
```

- Now go to “default-rule-path” and set it’s value to “/etc/suricata/rules” and “rule-files” to “*.rules”
- Save the file and restart the suricata by the command “sudo systemctl restart suricata” .
- Now we need to call the suricata file in the ossec.conf file.
- Navigate to ossec.conf and add the following tags under the log file format


```
<localfile>
  <log_format>json</log_format>
  <location>/var/log/suricata/eve.json</location>
</localfile>
```
- Save the file and restart the agent by command “systemctl restart wazuh-agent”
- Now to test this we need to run syn scan against the agent’s ip using nmap and observe it on the wazuh manager as shown:

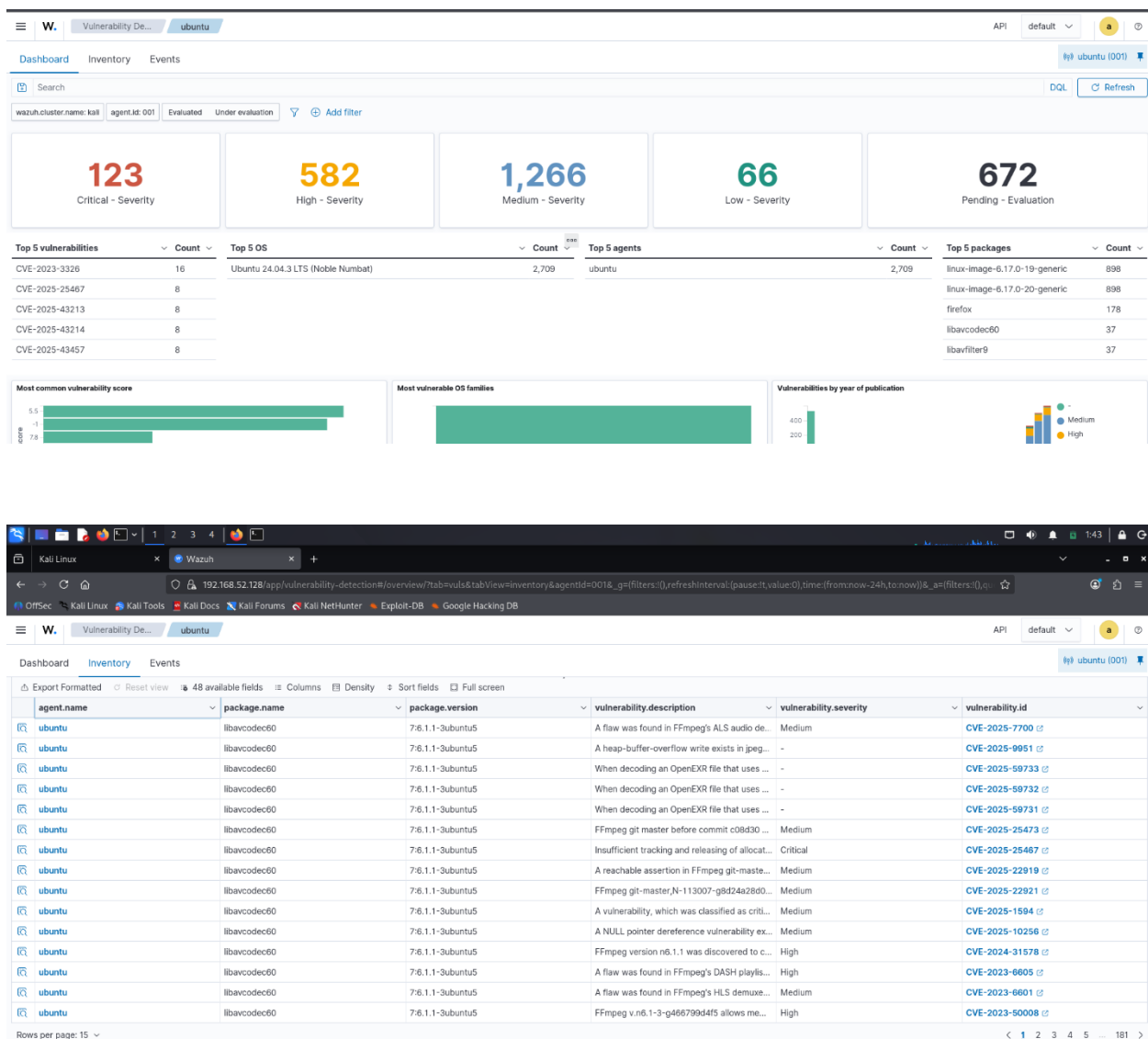


- we have got the alert.
- This completes our suricata ids for network monitoring.

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5. Detecting vulnerabilities

- To enable the vulnerability detection, we've to make changes in configuration file in the wazuh manager.
- Navigate to ossec.conf file by entering the command "nano /var/ossec/etc/ossec.conf"
- Go to the vulnerability-detector/detection section and enable it.
- Save the file.
- Now on the wazuh manager in the vulnerability detection section of the ubuntu agent we can see the vulnerabilities has been detected with severity levels, CVE id, CVE scores etc.
- This completes our vulnerability detection.

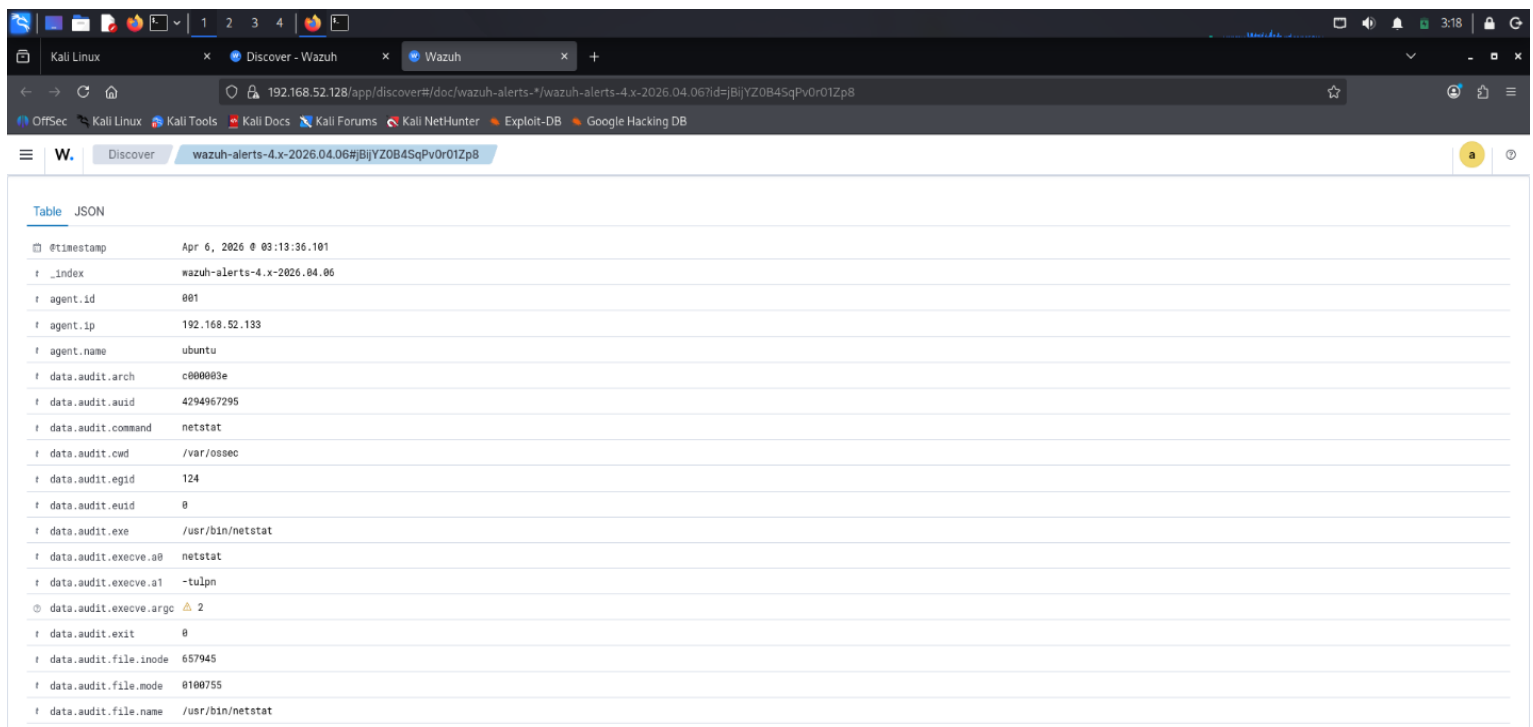


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6. Detecting Execution of Malicious Commands

- Install the “auditd” on the agent system by using the command “sudo apt install -y auditd”
- Navigate to ossec.conf file by using the command “nano /var/ossec/etc/ossec.conf” and add the following attribute:
“ <localfile>
 <log_format>audit</log_format>
 <location>/var/log/audit/audit.log</location>
 </localfile> “
- Save the file and restart the agent
- Open audit.rules file by using the command “nano /etc/audit/audit.rules” and write the following:
-a exit,always -F euid=0 -F arch=b64 -S execve -k audit-wazuh-c
-a exit,always -F euid=0 -F arch=b32 -S execve -k audit-wazuh-c
- Save the file and restart the auditd rules using the command “auditctl -R /etc/audit/audit.rules”
- Now run “netstat” on the agent terminal and on the wazuh dashboard we’ve got the alert.



The screenshot shows a Kali Linux terminal window at the top with the command `netstat` being executed. Below the terminal is a web browser window displaying the Wazuh dashboard. The dashboard shows a table of alerts for the command `netstat` executed on an Ubuntu agent.

Table JSON	
@timestamp	Apr 6, 2026 @ 03:13:36.101
_index	wazuh-alerts-4.x-2026.04.06
agent.id	001
agent.ip	192.168.52.133
agent.name	ubuntu
data.audit.arch	c000003e
data.audit.auid	4294967295
data.audit.command	netstat
data.audit.cwd	/var/ossec
data.audit.egid	124
data.audit.euid	0
data.audit.exe	/usr/bin/netstat
data.audit.execve.a0	netstat
data.audit.execve.a1	-tulpn
data.audit.execve.args	2
data.audit.exit	0
data.audit.file.inode	657945
data.audit.file.mode	0100755
data.audit.file.name	/usr/bin/netstat

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The screenshot shows a Kali Linux desktop environment with a browser window open to the Wazuh Discover interface. The browser tabs include 'Kali Linux', 'Discover - Wazuh', and 'Wazuh'. The address bar shows the URL: `192.168.52.128/app/discover#/doc/wazuh-alerts-*/wazuh-alerts-4.x-2026.04.06?id=BijYZ0B4SqPv0r01Zp8`. The Discover tab is active, displaying a list of alerts. The selected alert is expanded, showing the following details:

data.audit.type	SYSCALL
data.audit.uid	0
decoder.name	auditd
decoder.parent	auditd
full_log	>
id	1775459616.283027
input.type	log
location	/var/log/audit/audit.log
manager.name	kali
rule.description	Audit: Command: /usr/bin/netstat.
# rule.firedtimes	5
rule.gdpr	IV_30.1.g
rule.groups	audit, audit_command
rule.id	88792
# rule.level	3
rule.mail	false
timestamp	Apr 6, 2026 @ 03:13:36.101

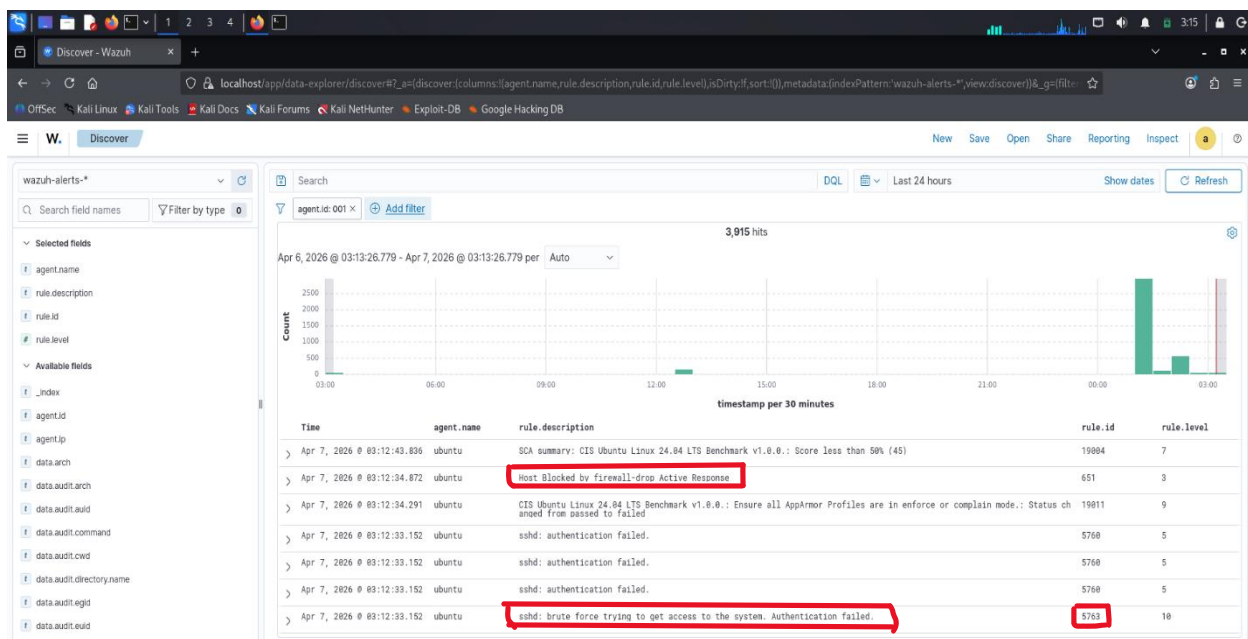
The 'full_log' field contains a detailed audit message: `type=SYSCALL msg=audit(1775459615.915:224): arch=c000003e syscall=59 success=yes exit=0 a0=581086d0cad8 a1=581086d0c0e0 a2=581086d0c8f8 a3=1 items=2 ppid=18419 pid=18420 auid=4294967295 uid=0 gid=124 euid=0 suid=0 fsuid=0 egid=124 sgd=124 fsgid=124 tty=(none) ses=4294967295 comm='netstat' exe='/usr/bin/netstat' subj=runconfined key='audit-wazuh-c' ARCH=x86_64 SYSCALL=execve AUID='unset' UID='root' GID='wazuh' EUID='root' SUID='root' FSUID='root' EGID='wazuh' SGID='wazuh' FSGID='wazuh' type=EXECVE msg=audit(1775459615.915:224): argc=2 a0='netstat' a1='-tulpn' type=CMD msg=audit(1775459615.915:224): cwd='/var/ossec' type=PATH msg=audit(1775459615.915:224): item=0 name='/usr/bin/netstat' inode=657945 dev=08:02 mode=0100755 ouid=0 ogid=0 rdev=00:00 nametype=NORMAL cap_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=00UID='root' OGID='root' type=PATH msg=audit(1775459615.915:224): item=1 name='/lib64/ld-linux-x86-64.so.2' inode=657831 dev=08:02 mode=0100755 ouid=0 ogid=0 rdev=00:00 nametype=NORMAL cap_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=00UID='root' OGID='root' type=PROCTITLE msg=audit(1775459615.915:224): proctitle=6E657473746174002074756C7065`

- This completes our detection of malicious command execution.

7. Detecting and Blocking SSH brute force attack

- On the wazuh manager navigate to ossec.conf file and add the following script to the <!--Active Response--> section:

```
<active-response>  
<command>firewall-drop</command>  
<location>local</location>  
<rules_id>5763</rules_id>  
<timeout>180</timeout>  
</active-response>
```
- Save the file and restart the wazuh manager
- Launch SSH attack using hydra for testing
- As we can see we've got the alert



Discover - Wazuh

localhost/app/discover/doc/wazuh-alerts-*wazuh-alerts-4.x-2026.04.07#id=EqzJZp0BjBwK9nJ4T7-0

Discover wazuh-alerts-4.x-2026.04.07#EqzJZp0BjBwK9nJ4T7-0

Table JSON

```
{
  "@timestamp": "2026-04-07T08:12:33.152Z",
  "_index": "wazuh-alerts-4.x-2026.04.07",
  "agent.id": "001",
  "agent.ip": "192.168.52.133",
  "agent.name": "ubuntu",
  "data.detuser": "owais",
  "data.scrip": "192.168.52.130",
  "data.srport": "48666",
  "decoder.name": "sshd",
  "decoder.parent": "sshd",
  "full_log": "Apr 07 07:11:31 owais-Vmware-Virtual-Platform sshd[14464]: Failed password for owais from 192.168.52.130 port 48666 ssh2",
  "id": "1775545953.684492",
  "input.type": "log",
  "location": "journal",
  "manager.name": "kali",
  "predecoder.hostname": "owais-Vmware-Virtual-Platform",
  "predecoder.program.name": "sshd",
  "predecoder.timestamp": "Apr 07 07:11:31",
  "previous_output": "Apr 07 07:11:31 owais-Vmware-Virtual-Platform sshd[14464]: Failed password for owais from 192.168.52.130 port 48666 ssh2\nApr 07 07:11:31 owais-Vmware-Virtual-Platform sshd[14465]: Failed password for owais from 192.168.52.130 port 48670 ssh2"
}
```

Screenshot taken

Discover wazuh-alerts-4.x-2026.04.07#EqzJZp0BjBwK9nJ4T7-0

```
{
  "previous_output": "Apr 07 07:11:31 owais-Vmware-Virtual-Platform sshd[14464]: Failed password for owais from 192.168.52.130 port 48666 ssh2\nApr 07 07:11:31 owais-Vmware-Virtual-Platform sshd[14465]: Failed password for owais from 192.168.52.130 port 48670 ssh2\nApr 07 07:11:31 owais-Vmware-Virtual-Platform sshd[14467]: Failed password for owais from 192.168.52.130 port 48690 ssh2\nApr 07 07:08:53 owais-Vmware-Virtual-Platform sshd[14481]: Failed password for owais from 192.168.52.130 port 34354 ssh2\nApr 07 07:08:53 owais-Vmware-Virtual-Platform sshd[14448]: Failed password for owais from 192.168.52.130 port 34304 ssh2\nApr 07 07:08:53 owais-Vmware-Virtual-Platform sshd[14490]: Failed password for owais from 192.168.52.130 port 34320 ssh2\nApr 07 07:08:53 owais-Vmware-Virtual-Platform sshd[14480]: Failed password for owais from 192.168.52.130 port 34399 ssh2",
  "rule.description": "sshd: brute force trying to get access to the system. Authentication failed.",
  "# rule.firetimes": "1",
  "# rule.frequency": "8",
  "rule.gdpr": "IV.35.7.d, IV.32.2",
  "rule.groups": "syslog, sshd, authentication_failures",
  "rule.hipaa": "164.312.b",
  "rule.id": "5763",
  "# rule.level": "10",
  "rule.mail": "false",
  "rule.mitre.id": "T1118",
  "rule.mitre.tactic": "Credential Access",
  "rule.mitre.technique": "Brute Force",
  "rule.nist_800_53": "SI.4, AU.14, AC.7",
  "rule.pci_dss": "11.4, 10.2.4, 10.2.5",
  "rule.tao": "C06.1, C06.8, C07.2, C07.3",
  "@timestamp": "2026-04-07T08:12:33.152Z"
}
```

Discover wazuh-alerts-4.x-2026.04.07#EqzJZp0BjBwK9nJ4T7-0

```
{
  "data.version": "1",
  "decoder.name": "ar_log_json",
  "decoder.parent": "ar_log_json",
  "full_log": "2026/04/07 12:42:33 active-response/bin/firewall-drop: {\"version\":\"1\", \"origin\":{\"name\":\"node01\", \"module\":\"wazuh-execd\", \"command\":\"add\", \"parameters\":{\"extra_args\":[]}, \"alert\":{\"timestamp\":\"2026-04-07T08:12:33.152-0400\", \"rule\":{\"level\":\"10\", \"description\":\"sshd: brute force trying to get access to the system. Authentication failed.\"}, \"id\":\"5763\", \"mitre\":{\"id\":\"T1118\", \"tactic\":\"Credential Access\", \"technique\":\"Brute Force\"}}, \"frequency\":\"8\", \"firetimes\":\"1\", \"mail\":\"false\", \"groups\":{\"syslog\", \"sshd\", \"authentication_failures\"}, \"gdpr\":{\"IV.35.7.d\", \"IV.32.2\"}, \"hipaa\":{\"164.312.b\"}, \"nist_800_53\":{\"SI.4\", \"AU.14\", \"AC.7\"}, \"pci_dss\":{\"11.4\", \"10.2.4\", \"10.2.5\"}, \"tao\":{\"C06.1\", \"C06.8\", \"C07.2\", \"C07.3\"}}, \"agent\":{\"id\":\"001\", \"name\":\"ubuntu\", \"ip\":\"192.168.52.133\"}, \"manager\":{\"name\":\"kali\", \"id\":\"1775545953.684107\", \"previous_output\":\"Apr 07 07:11:31 owais-Vmware-Virtual-Platform sshd[14464]: Failed password for owais from 192.168.52.130 port 48666 ssh2\\nApr 07 07:11:31 owais-Vmware-Virtual-Platform sshd[14465]: Failed password for owais from 192.168.52.130 port 48670 ssh2\\nApr 07 07:11:31 owais-Vmware-Virtual-Platform sshd[14467]: Failed password for owais from 192.168.52.130 port 48690 ssh2\\nApr 07 07:08:53 owais-Vmware-Virtual-Platform sshd[14481]: Failed password for owais from 192.168.52.130 port 34354 ssh2\\nApr 07 07:08:53 owais-Vmware-Virtual-Platform sshd[14448]: Failed password for owais from 192.168.52.130 port 34304 ssh2\\nApr 07 07:08:53 owais-Vmware-Virtual-Platform sshd[14490]: Failed password for owais from 192.168.52.130 port 34320 ssh2\\nApr 07 07:08:53 owais-Vmware-Virtual-Platform sshd[14480]: Failed password for owais from 192.168.52.130 port 34399 ssh2\""}, \"timestamp\":\"2026-04-07T08:12:33.152Z\"}",
  "id": "1775545954.6849576",
  "input.type": "log",
  "location": "/var/ossec/logs/active-responses.log",
  "manager.name": "kali",
  "rule.description": "Host Blocked by firewall-drop Active Response",
  "# rule.firetimes": "1",
  "rule.gdpr": "IV.35.7.d",
  "rule.pgpg13": "4.13",
  "rule.groups": "ossec, active_response",
  "rule.id": "651",
  "# rule.level": "3",
  "rule.mail": "false",
  "rule.nist_800_53": "SI.4",
  "rule.pci_dss": "11.4"
}
```

Name: OWAIS YOUSUF
Date: 09-04-2026

8. Detecting malicious files using Virustotal

- On the wazuh manager go to management>Rules and search for “local”
- Edit the local_rules.xml files and add the following:

```
<rule id="100200" level="7">  
  <if_sid>550</if_sid>  
  <field name="file">/root</field>  
  <description>File added to /root directory.</description>  
</rule>
```



```
1 <!-- Local rules -->  
2  
3 <!-- Modify it at your will. -->  
4 <!-- Copyright (C) 2015, Wazuh Inc. -->  
5  
6 <!-- Example -->  
7 <group name="local,syslog,sshd,">  
8  
9 <!--  
10 Dec 10 01:02:02 host sshd[1234]: Failed none for root from 1.1.1.1 port 1066 ssh2  
11 -->  
12 <rule id="100001" level="5">  
13 <if_sid>5716</if_sid>  
14 <srcip>1.1.1.1</srcip>  
15 <description>sshd: authentication failed from IP 1.1.1.1.</description>  
16 <group>authentication_failed,pci_dss_10.2.4,pci_dss_10.2.5,</group>  
17 </rule>  
18  
19 <rule id="100200" level="7">  
20 <if_sid>550</if_sid>  
21 <field name="file">/root</field>  
22 <description>File added to /root directory.</description>  
23 </rule>  
24  
25 </group>  
26
```

- Save the file and reload
- Now go to ossec.conf file in wazuh manager the add the following integration attribute after the cluster attribute:

```
<integration>  
  <name>virustotal</name>  
  <api_key>“Your api key ”</api_key>  
  <rule_id>100200,100201</rule_id>  
  <alert_format>json</alert_format>  
</integration>
```

Note: Api key can be found on virustotal under profile section

- Save the file
- On the agent terminal run the following :
“curl -Lo /root/eicar.com https://secure.eicar.org/eicar.com && sudo ls -lah /root/eicar.com” it will download the eicar file
- On the wazuh manager we’ve got the alert
- This completes our malicious file detection using virustotal.

Discover - Wazuh

192.168.52.128/app/discover#/doc/wazuh-alerts-*?wazuh-alerts-4.x-2026.04.08?Id=Mjza50Bv42LUQzW4bEs

Discover wazuh-alerts-4.x-2026.04.08#Mjza50Bv42LUQzW4bEs

Table JSON

@timestamp	Apr 8, 2026 @ 03:17:16.006
_index	wazuh-alerts-4.x-2026.04.08
agent.id	001
agent.ip	192.168.02.133
agent.name	ubuntu
data.integration	virusotal
data.virusotal.found	1
data.virusotal.malicious	1
data.virusotal.permalink	https://www.virustotal.com/gui/file/275a821bbf6489e54d471899f7db9d1663fc959ec2f62a2c4538aabf651fd8f/detection/f-275a821bbf6489e54d471899f7db9d1663fc959ec2f62a2c4538aabf651fd8f-1775632255
data.virusotal.positives	66
data.virusotal.scan_date	2026-04-08 07:10:55
data.virusotal.sha1	3395856ce81f2b7382dee72602f79b642f14149
data.virusotal.source.alert_id	1775632631.45129
data.virusotal.source.file	/root/.eicar.com
data.virusotal.source.md5	44d88612feaba8f36de82e1278abb02f
data.virusotal.source.sha1	3395856ce81f2b7382dee72602f79b642f14149
data.virusotal.total	68
decoder.name	json
id	1775632636.47007

Screenshot taken

Discover wazuh-alerts-4.x-2026.04.08#Mjza50Bv42LUQzW4bEs

data.virusotal.source.md5	44d88612feaba8f36de82e1278abb02f
data.virusotal.source.sha1	3395856ce81f2b7382dee72602f79b642f14149
data.virusotal.total	68
decoder.name	json
id	1775632636.47007
input.type	log
location	virusotal
manager.name	kali
rule.description	VirusTotal: Alert - /root/.eicar.com - 66 engines detected this file
# rule.firedtimes	1
rule.gdpr	IV_35.7.d
rule.groups	virusotal
rule.id	87105
# rule.level	12
rule.mali	true
rule.mitre.id	T1203
rule.mitre.tactic	Execution
rule.mitre.technique	Exploitation for Client Execution
rule.pci_dss	10.6.1, 11.4
@timestamp	Apr 8, 2026 @ 03:17:16.006

Name: OWAIS YOUSUF

Date: 09-04-2026

Results

Results

- Wazuh successfully detected file changes, suspicious commands, and system vulnerabilities in real time.
 - Integration with Suricata identified network-based attacks like port scanning and malicious traffic.
 - SSH brute-force attacks were detected and automatically blocked using active response.
 - Suspicious files were analyzed using VirusTotal, generating alerts for malicious files.
 - All events were centralized and visualized in the Wazuh dashboard.
-

Conclusion

- The project successfully implemented a **multi-layered security monitoring system** using Wazuh.
- It effectively detected and responded to different types of cyber threats.
- Integration with Suricata and VirusTotal enhanced detection accuracy.
- Overall, the system is a **practical and scalable solution** for improving cybersecurity in real-world environments.

References

- Wazuh official documentation: [Wazuh documentation](#)
- Suricata documentation: [Suricata-IDS-Home-Lab/installing-suricata.md at main · Oxrajneesh/Suricata-IDS-Home-Lab](#)
- VirusTotal official website: [VirusTotal - My API Key](#)
- Ubuntu documentation: [Ubuntu documentation](#)